

Outer correctors, Newton iterations and the Newmark scheme

1 October 2026 · Colm Mc Alister · beamFoam colm_6D0F_solver , moorFV feature-beamFoamCoupling-colm Companion to monolithicCouplingPlan.pdf (Phase 4). Every statement about behaviour is taken from the code cited or from the logs of the rigidBodyAndBeam_* tutorial runs.

In short. The PIMPLE outer correctors repeat the whole fluid–body–beam update inside a time step.

- **Loop framework.** Each corrector is one pass of a relaxed fixed-point iteration towards the implicit Newmark solution. With one corrector the beam force lags the body by a step. That makes the body's Newmark scheme explicit, first order and unstable at every time step (in a linear model).
- **Monolithic framework.** Newmark is already implicit in the beam force. Outer correctors only update the explicit fluid force, so one corrector is stable and second-order accurate.

Each extra corrector costs a full fluid solve plus 2–3 beam Newton iterations. In these cases the fluid solve dominates the run time.

1. Two loops called “outer corrector”

The name is used for two different loops, so it helps to separate them first:

Loop	Set by	What it repeats
PIMPLE outer corrector	nOuterCorrectors in system/fvSolution (PIMPLE)	The whole interFoam time step: mesh motion (body and beam), then α , U and p. This document is about this loop.
beamFoam Newton iteration	nCorrectors 200 (the maximum) in constant/beam/beamProperties	One Newton–Raphson iteration of the beam (or beam + body) system. The counter is called iOuterCorr() in the code, but it is a Newton iteration. The log line N: Converged reports N of them for one beam solve.

```
for each time step n → n+1
  for each PIMPLE outer corrector k = 1 ... nOuterCorrectors
    if (k == 1 or moveMeshOuterCorrectors) mesh.update()      ← body + beam
    moorFV motion solver: fluid force on body, then solver.solve(firstIter = (k ==
1))
      beamFoam evolve(): Newton iterations until converged    ← "N: Converged"
    solve  $\alpha$ , U, p
```

interFoam.C:112 (if (pimple.firstIter() || moveMeshOuterCorrectors) mesh.update(););
sixDoFRigidBodyMotionFvBeamSolver.C:212–268

moveMeshOuterCorrectors yes **is needed for any coupling iteration.** Without it, correctors 2 ... N repeat only the fluid solve. The body and beam are updated once per step whatever nOuterCorrectors is. The _nOuter8 cases set both.

2. What one outer corrector does in each framework

Start of a time step

On the first call in a time step, moorFV stores the start-of-step body state (`motion_.newTime() : motionState0_ = motionState_`). Every later corrector updates the body *from that stored state* (x_n, v_n, a_n), not from the previous corrector. In the monolithic path, beamFoam's `rigidBodyEnd::newTimeStep()` does the same once per time index. So repeating a corrector re-solves the same step with updated forces; it never advances time twice.

Loop framework (`FvBeamNewmark`), corrector k

1. **Beam solve with the end prescribed.** `updateAcceleration` calls the `finiteVolumeBeam` restraint before the body is moved. The beam end is set to the body's *current* position: on corrector 1 that is the end of the last step, x_n ; on corrector k it is $x^{(k-1)}$. beamFoam then solves the beam at time t_{n+1} .

`finiteVolumeBeam.C:277–286, 349`

2. **Acceleration with relaxation.** $a^* = (F_{\text{fluid}} + m g + F_{\text{beam}})/m$, then $a^{(k)} = 0.5 a^* + 0.5 a^{(k-1)}$ (`accelerationRelaxation 0.5`). `sixDoFRigidBodyMotionFvBeam.C:306–333`

3. **Newmark update** from the stored start-of-step state, with $\beta = 1/4$ and $\gamma = 1/2$:

$$v = v_n + \Delta t[\gamma a + (1-\gamma) a_n], \quad x = x_n + \Delta t v_n + \Delta t^2[\beta a + (\frac{1}{2}-\beta) a_n]$$

`FvBeamNewmark.C:74–115`

Relaxation on the first corrector of each step. The relaxation is skipped only on the first call of the whole run (a `static bool first`). On corrector 1 of every later step, $a^{(0)}$ is the previous step's acceleration a_n . With one corrector the body therefore uses $a = 0.5 a^* + 0.5 a_n$: half of the new acceleration and half of the old. This is a lag on top of the lagged beam force. Resetting it on `firstIter` would be a one-line change in moorFV; it hasn't been made.

Monolithic framework (`beamFoamCoupled`), corrector k

1. **Explicit applied load.** The fluid force and moment, the weight and any other restraints are taken at the current state.
2. **One Newton solve of beam + body.** `solveMonolithic` runs beamFoam's Newton iteration on the beam equations and the body's Newmark residual $R = m a(x) + c v(x) - F_{\text{ext}} - m g + Q_b(x)$ together, in one BlockEigen system. The Newmark relations are applied from the stored start-of-step state, so the beam force is implicit. The Newton iteration starts from the previous corrector's solution.
3. **Accept the state.** The converged state is copied into the motion. Nothing is recomputed or relaxed.

3. Effect on the Newton iterations

From the base tutorial logs (88–89 time steps to $t = 0.25$ s; air only, so the fluid force is small):

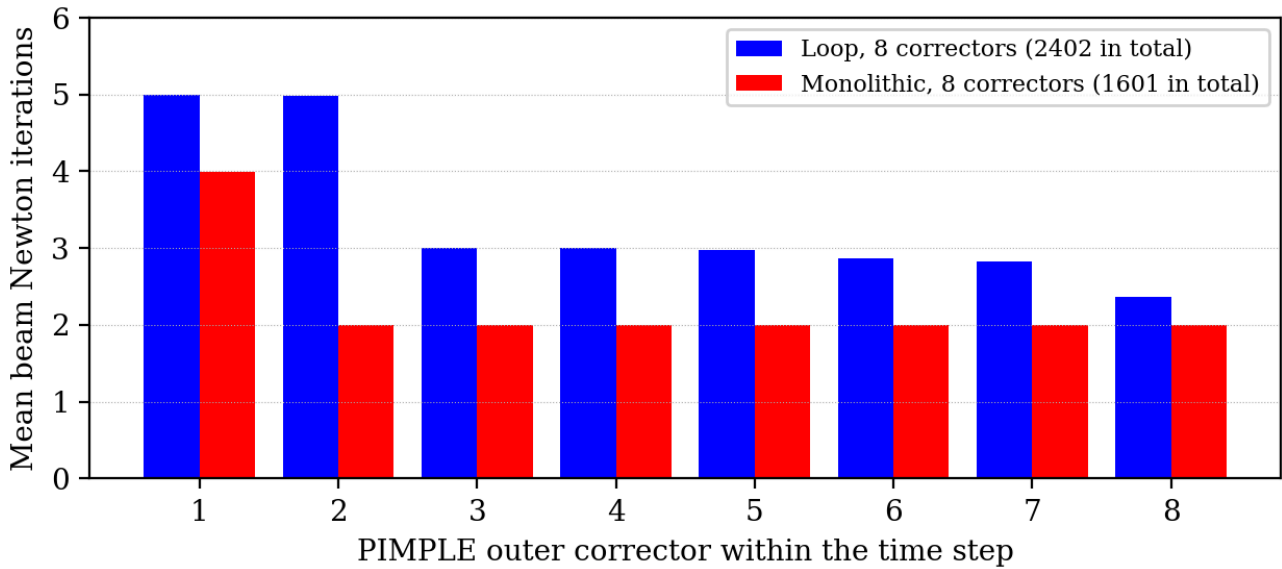


Figure 1. Mean beam Newton iterations for each PIMPLE outer corrector, `rigidBodyAndBeam_loop_nOuter8` and `rigidBodyAndBeam_beamFoamCoupled_nOuter8`.

Case	Beam solves	Newton iterations	Per step	Run time (s)
Loop, 1 corrector	88	439	5.0	359
Monolithic, 1 corrector	88	351	4.0	359
Loop, 8 correctors	712	2402	27.0	2219
Monolithic, 8 correctors	712	1601	18.0	2219

- **Loop.**
 - Corrector 1 takes about 5 iterations: the beam is advanced to t_{n+1} with its end still at x_n .
 - Corrector 2 also takes about 5: the end jumps by the whole step's body displacement.
 - Correctors 3–8 take about 3: the end keeps moving, by an amount that halves each corrector (the relaxation factor), so every beam solve has a new boundary value to absorb.
- **Monolithic.** Corrector 1 takes 4 iterations. Correctors 2–8 take 2 each, the fewest any beam solve takes in these runs: only the small fluid force has changed, and the iteration starts from the converged state.
- **Cost.** The run times are identical step by step between the frameworks: 359 s and 359 s, 2219 s and 2219 s. The 60-cell beam solve is negligible next to the fluid solve, so in this case the cost of an outer corrector is the fluid. The 6.2× saving reported in part 1 (monolithic with 1 corrector against the loop with 8) comes from needing fewer correctors, not from a faster beam solve. The Newton-iteration saving (1.25× with 1 corrector, 1.5× with 8) would matter only when the beam is a large share of the cost, for example with many segments or several lines.

4. Effect on the Newmark- β scheme

Newmark with $\beta = 1/4$ and $\gamma = 1/2$ (average acceleration, the trapezoidal rule) has three properties:

- it is unconditionally stable;
- it is second-order accurate;
- it adds no numerical damping.

These hold *only* when the end-of-step acceleration a_{n+1} is consistent with the end-of-step position x_{n+1} , that is, when the scheme is solved implicitly for every stiff force. The outer correctors control whether this is true for the beam force.

A model problem

The line seen by the body is modelled as a linear spring k on a mass m . The line's own mass is about 1.5×10^{-4} kg against a 15 kg body, so the spring needs no mass. The model is then advanced with exactly the update order in §2, giving a 3×3 amplification matrix on (x, v, a) . The important number is $\Omega = \omega_n \Delta t$, where $\omega_n = \sqrt{k/m}$.

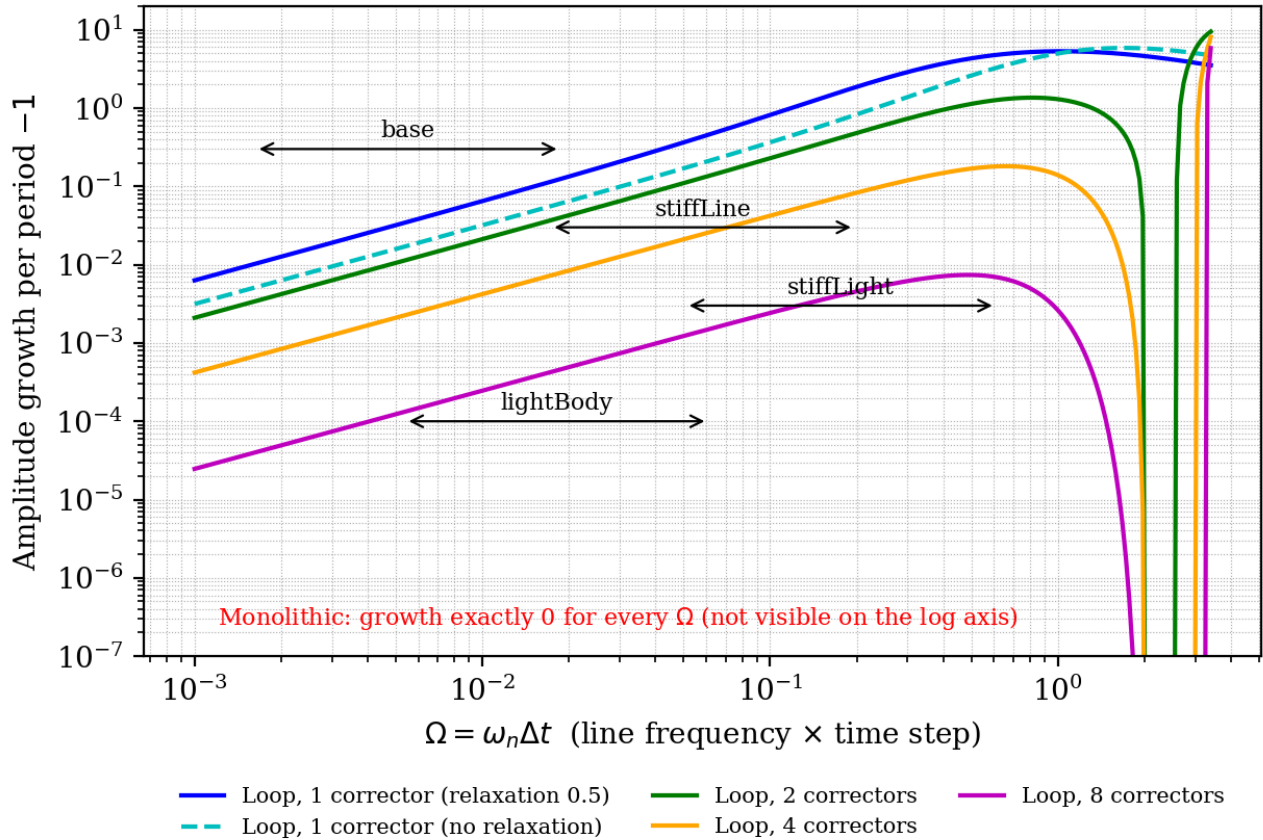


Figure 2. Growth of the line-body oscillation per period in the model, against $\Omega = \omega_n \Delta t$. The loop uses relaxation 0.5 unless stated. Curves that leave through the bottom are damped there, not growing. Arrows: the range of Ω in each case, from the starting $\Delta t = 0.01$ s to the Courant-limited $\Delta t \approx 8.5 \times 10^{-4}$ s, using $k = EA/L$ for the axial line stiffness.

Scheme	Stability	Accuracy
Loop, 1 corrector	Unstable for every Ω. The beam force lags the displacement, which acts as negative damping. Growth per period is about $e^{2\pi\Omega}$ with relaxation 0.5 and $e^{\pi\Omega}$ without it: 13% per period at $\Omega = 0.02$, rising to about $\times 6$ at $\Omega \approx 1$. Phase 0 measured +15% per period for one-pass partitioned coupling, which matches.	First order: the local error falls 4 $\times$ when Δt is halved.
Loop, K correctors	Fixed-point iteration towards the implicit solution, with contraction factor $\rho = 1 - \omega_r(1 + \beta\Omega^2) $, where ω_r is the relaxation factor. With $\omega_r = 0.5$ and small Ω , $\rho \approx 0.5$: the error halves per corrector, as seen in the part 1 runs. The iteration diverges within the step if $\Omega > 2\sqrt{3} \approx 3.5$ ($\omega_r = 0.5$) or $\Omega > 2$ (no relaxation). The leftover growth falls about 2 $\times$ per corrector; 7–11 correctors are needed for growth below 0.1% per period.	Tends to second order as K increases, but stays first order with a small coefficient for any finite K.
Monolithic, any number of correctors	$\lambda = 1$ exactly for every Ω: no growth and no numerical damping from the beam–body coupling.	Second order: the local error falls 8 $\times$ when Δt is halved.

What the extra monolithic correctors do

The only explicit term left in the monolithic path is the fluid force (and any other restraint). Each extra corrector re-solves the step from the stored start-of-step state, using the latest fluid force. With converged correctors, the body's Newmark scheme is then implicit in *all* forces. In air the fluid force barely changes between correctors, which is why one corrector already agrees with eight to about 3×10^{-4} of the peak displacement (Phase 4, part 1). In water, a light body with a large added mass would need more correctors, as any partitioned fluid–structure scheme does; that is separate from the beam coupling.

The beam's own time integration

The beam has its own scheme. This tutorial uses `d2dt2Schemes default Euler` in `system/beam/fvSchemes`, so the beam's inertia is integrated with a first-order, dissipative Euler scheme, while the body uses Newmark $\beta = 1/4$, $\gamma = 1/2$. `beamFoam`'s Newmark predictor and coefficients apply only when the beam's scheme is `Newmark`. The line's mass is tiny here, so this barely affects the body. With a heavy line it would add numerical damping to the line's own waves, whatever the coupling.

5. What this means for the tutorial and the stress tests

- **Base case.** $\Omega \approx 0.002$ – 0.02 and the line–body period (about 3.2 s) is far longer than the 0.25 s run. The loop's growth of a few per cent per period has no time to appear, which is why loop with 1 corrector differs from the converged loop by only 5×10^{-4} of the peak displacement.
- **Stress tests.**
 - `stiffLine` ($E \times 100$): $\Omega \approx 0.02$ – 0.2 , period about 0.32 s.
 - `stiffLight` (also $m \div 10$): $\Omega \approx 0.05$ – 0.6 , period about 0.1 s, so about 2.5 periods in the run.

The model predicts that the loop with 1 corrector will grow by tens of per cent to several times per period in these cases, while the monolithic solver stays neutral.

Measured (`tutorials/rigidBodyAndBeam_stress` ; maximum force error against monolithic with 8 correctors, as a fraction of its peak):

Variant	Loop, 1	Monolithic, 1	Loop, 8	Growth of the loop-1 error over the run
stiffLine	4.3e-2	4.6e-4	1.9e-4	×39
stiffLight	1.2e-1	3.4e-3	2.2e-4	×15
stiffLargeDt (maxCo 1)	1.5e-1	5.1e-4	6.2e-4	×115

No run diverged within 0.25 s. The loop's error with 1 corrector grows steadily, as the negative damping predicts, but more slowly than the model's rate. The real system has damping the model leaves out: the beam's Euler time integration, the fluid and the nonlinear line. Treat the model as an upper bound on the growth rate. Monolithic with 1 corrector stays within 0.34%.

- **Settings.**

- **Monolithic:** 1 outer corrector is enough when the fluid force is weakly coupled. Use 2–3 (with `moveMeshOuterCorrectors yes`) when fluid added mass matters. Correctors after the first cost a fluid solve plus 2 Newton iterations.
- **Loop:** use enough correctors to converge the fixed point (about 8 with relaxation 0.5), and keep Ω well below $2\sqrt{3}$.

Model script: amplification matrices for the update order in §2; growth per period = $\rho^{2\pi/\Omega}$. Newton counts are from `N:`
`Converged` lines grouped by `PIMPLE: iteration k`; run times are the final `ExecutionTime`.